

Amirreza Khoshbakht

Curriculum Vitae

Contact

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- 📄 GoogleScholar Amirreza Khoshbakht

Professional Skills

- Deep Learning
- Computer Vision
- Uncertainty Quantification
- Domain Generalization
- Remote Sensing

Honors and Awards

- **IEEE SIU2026 Student Paper Award**
Best Student Paper Award in Image Processing and Computer Vision Track
Jul 2026
- **IEEE SIU2025 Student Paper Award**
Second Best Student Paper Award in Image Processing and Computer Vision Track
Jun 2025
- **Data Science Award**
Bilkent Senior Design Project Awards for the WePrep Project
Jun 2024
- **Bilkentpreneurs Challenge**
1st team of the Bilkentpreneurs Challenge 2024 with the WePrep Project
Jun 2024

Teaching Experience

- **CS515: Deep Learning**
Sabanci University
Graduate level course, Weekly tutorial sessions, homework preparation, and exam grading.
Feb 2026 - present
- **CS415: Introduction to Deep Learning**
Sabanci University
Lab Sessions, office hours, homework preparation, and exam grading.
Sept 2025 - Jan 2026
- **CS201: Fundamentals of Programming**
Sabanci University
Weekly recitation sessions, office hours, homework preparation, and exam grading.
Sept 2024 - Jun 2025

Education

2024-
Present



Masters of Science

📍 Istanbul, Turkey

Sabanci University Faculty of Engineering,
Masters of Data Science

Ongoing Thesis Area: Open-Set Recognition via Uncertainty Quantification

GPA: 3.9/4

2020-2024



Bachelor of Science

📍 Ankara, Turkey

Bilkent University Faculty of Engineering,
Bachelor of Science in Computer Engineering

2023-2023



Erasmus Exchange Student

📍 Roskilde, Denmark

Roskilde University Faculty of Engineering,
Erasmus Exchange Student (6 Months)

Publications

Journal Articles

GRSL

Apr. 2026

"Hyperbolic Prototype Learning for Open-Set Joint HSI-LiDAR Classification", **Khoshbakht A**, Aptoula E, *IEEE Geoscience and Remote Sensing Letters*, [doi](#) 10.1109/LGRS.2026.3688851

Under

Review

2026

"Open-Set Domain Generalization through Spectral-Spatial Uncertainty Disentanglement for Hyperspectral Image Classification", **Khoshbakht A**, Aptoula E, *Under Review in Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, [doi](#) 10.48550/arXiv.2506.09460

Conference & Workshop Papers

IEEE SIU

Jul. 2026

"Spectral-Spatial Open-Set Recognition With Adaptive Contrastive Semantic Reconstruction for Hyperspectral Imagery", **Khoshbakht A**, Aptoula E, *Signal Processing and Communications Applications Conference; Istanbul, Türkiye*, [doi](#) TBA

CVPR

Workshop

Jun. 2026

"Hierarchical Uncertainty-Aware Deep Simplex Classification for Open-Set Medical Image Recognition", **Khoshbakht A**, Aptoula E, *PHAROS AI Factory for Medical Imaging & Healthcare CVPR Workshop; Denver, Colorado, USA*, [doi](#) TBA

IEEE

Whispers

Nov. 2025

"Open-Set Domain Generalization for Hyperspectral Image Classification via Evidential Uncertainty Quantification", **Khoshbakht A**, Aptoula E, *Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing; Barcelona, Spain*, [doi](#) 10.1109/WHISPERS69515.2025.11501598

IEEE

Whispers

Nov. 2025

"GEOSMAVIT: Geographical-Context Spectral Invariant Multi-Scale Adaptive Vision-Language Transformer", **Abu Bakar A**, **Khoshbakht A**, Aptoula E, *Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing; Barcelona, Spain*, [doi](#) 10.1109/WHISPERS69515.2025.11501561

IEEE SIU

Jun. 2025

"Semantically Guided Gradient Matching for Open-Set Domain Generalization", **Khoshbakht A**, Aptoula E, *Signal Processing and Communications Applications Conference; Istanbul, Türkiye*, [doi](#) 10.1109/SIU66497.2025.11112269

Awarded Scholarships

- TUBITAK Scholarship**
Granted a full tuition waiver and a monthly stipend while working as a researcher on a TUBITAK-funded project.
Jul 2025
- Sabanci University Scholarship**
Received a full tuition waiver along with a monthly stipend for serving as a Teaching Assistant
Sept 2024
- Bilkent University Scholarship**
100% Academic success scholarship
Sept 2020

Languages

- English - Professional Working Proficiency
- Persian - Native Languagey
- Azerbaijani - Native Language
- Turkish - Limited Working Proficiency

Selected Courses

- Machine Leaning:A (4/4)
- Deep Learning: A- (3.7/4)
- Scalable Learning Systems: A- (3.7/4)
- Directed Research: A (4/4)

Technical Skills

Python

C++

Java

Numpy

Pytorch

OpenCV

MySQL

Spring Framework

RESTful APIs

React.JS

References

- Dr. Erchan Aptoula** (Supervisor)
Professor, Faculty of Engineering and Natural Sciences, Sabanci University, İstanbul, Türkiye
Email: erchan.aptoula@sabanciuniv.edu
- Prof. Orçun Çetin**
Professor, Faculty of Engineering and Natural Sciences, Sabanci University, İstanbul, Türkiye
Email:orcun.cetin@sabanciuniv.edu

Projects

- Computer Vision and Pattern Analysis(VPA) Laboratory**
Sabanci University | Sept 2024 - Present
Developed novel domain generalization methods for remote sensing image analysis enabling robust performance across diverse sensors and domains. Created open-set domain generalization frameworks for hyperspectral image classification using evidential deep learning and spectrum-invariant frequency disentanglement, achieving state-of-the-art performance without target domain access. Designed semantically-guided gradient matching techniques that incorporate continuous class relationships into meta-learning frameworks, outperforming existing methods on multiple benchmarks including PACS, Houston, and Pavia datasets.
- WePrep** - Python, React, AWS Lambda, REST APIs
Bilkent University | Sept 2023 - Jun 2024
WePrep simulates tailored mock interviews and presentations through a virtual voice, analyzing your performance by evaluating factors like facial expressions, body language, and answer quality. The application provides instant, unbiased feedback, along with personalized recommendations, creating an efficient pathway for your skill enhancement.
- PRAssist** - Node.JS, Probot Framework
Bilkent University | Feb 2024 - Jun 2024
PRAssist automates GitHub pull request management using an LLM to generate titles and descriptions from file diffs, and suggests reviewers based on past contributions and code expertise. This streamlines the process, saving developers time while improving consistency and team collaboration. Github
- Defense Against Natural Adversarial Examples** - Python
Bilkent University | Feb 2024 - Jun 2024
This project presents a novel DNN defense mechanism against natural adversarial examples in ImageNet using Salient Feature Extraction (SFE) via coupled GANs. The method distinguishes robust, human-aligned salient features from misleading trivial features, significantly improving model accuracy and robustness with substantial gains on challenging datasets like ImageNet-A. This addresses the critical need for reliable defenses against naturally occurring adversarial perturbations in real-world applications.

Work Experience

- Software Engineering Intern
YD Software and Technology, Ankara, Turkey | Jun. 2023 – Aug 2023
 - ontributed to the development and enhancement of the company's **Mentally Video Conferencing** application.
 - Implemented real-time communication for the application utilizing **Google's Webrtc**.
 - Developed the backend of the application using **Node.JS**.
 - Implemented the frontend of the application using **React.JS**.
 - Facilitated the connection between the backend and frontend using **Mediasoup**.
- Java Developer Intern
Layermark, Remote | Jun. 2022 – Aug 2022
 - Improved company's **Emergency Management System(EMS)** application.
 - RESTful APIs** were used to implement several services.
 - Twilio SendGrid API** was used to implement a email/sms sender service for the MMS project.
 - PostgreSQL, MySQL, and Elasticsearch** queries and triggers were used to communicate and implement services for the EMS project.
 - Java SpringBoot framework** was used to implement services.